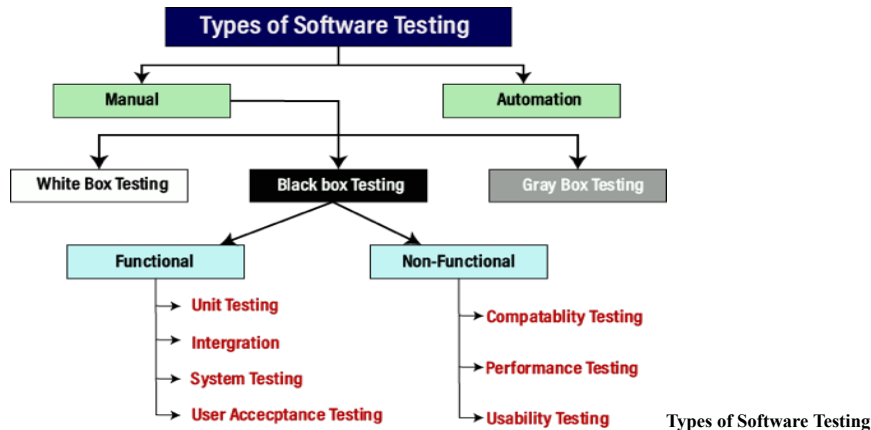


## INTRODUCTION

Software testing is a process of identifying the correctness of software by considering its all attributes (Reliability, Scalability, Portability, Re- usability, Usability) and evaluating the execution of software components to find the software bugs or errors or defects. The process of software testing aims not only at finding faults in the existing software but also to improve the software in terms of efficiency, accuracy and usability.



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### Manual testing

The process of checking the functionality of an application as per the customer needs without taking any help of automation tools is known as manual testing. While performing the manual testing on any application, we do not need any specific knowledge of any testing tool.

Manual testing can be further divided into three types of testing,

White box testing  
Black box testing  
Gray box testing

### Automation testing

Automation testing is a process of converting any manual test cases into the test scripts with the help of automation tools, or any programming language is known as automation testing.

With the help of automation testing, we can enhance the speed of our test execution.

We do not require any human efforts.

We need to write a test script and execute those scripts.

## Functional Testing

- It is a type of software testing that validates the software systems against the functional requirements.
- It is performed to check whether the application is working as per the software's functional requirements or not.
- Various types of functional testing are Unit testing, Integration testing, System testing and so on.

This testing is not concerned with the source code of the application. Each functionality of the software application is tested by providing appropriate test input, expecting the output, and comparing the actual output with the expected output.

### Purpose of Functional Testing

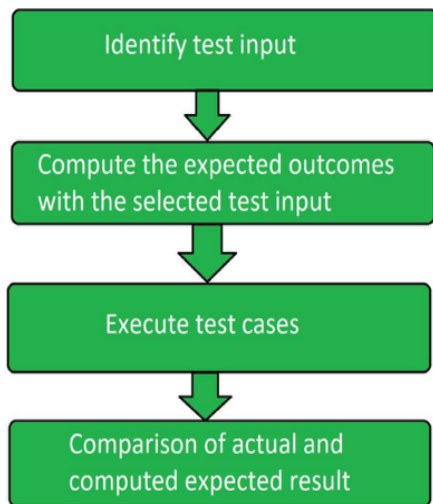
- Functional testing mainly involves black box testing and can be done manually or using automation.
- Test each function of the application: Functional testing tests each function of the application by providing the appropriate input and verifying the output against the functional requirements of the application.
- Test primary entry function: In functional testing, the tester tests each entry function of the application to check all the entry and exit points.
- Test flow of the GUI screen: In functional testing, the flow of the GUI screen is checked so that the user can navigate throughout the application.

### Goal of Functional Testing

- Basic Usability: Functional testing involves basic usability testing to check whether the user can freely navigate through the screens without any difficulty.
- Mainline functions: This involves testing the main features and functions of the application.
- Accessibility: This involves testing the accessibility of the system for the user.
- Error Conditions: Functional testing involves checking whether the appropriate error messages are being displayed or not in case of error conditions.

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### Process of Functional Testing



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#### Types of Functional Testing:

1. Unit Testing: Unit testing is the type of functional testing technique where the individual units or modules of the application are tested. It ensures that each module is working correctly.
2. Integration Testing: In Integration testing, combined individual units are tested as a group and expose the faults in the interaction between the integrated units.
3. Smoke Testing: Smoke testing is a type of functional testing technique where the basic functionality or feature of the application is tested as it ensures that the most important function works properly.
4. User Acceptance Testing: User acceptance testing is done by the client to certify that the system meets the requirements and works as intended. It is the final phase of testing before the product release.
5. Interface Testing: Interface testing is a type of software testing technique that checks the proper interaction between two different software systems.
6. System Testing: System testing is a type of software testing that is performed on the complete integrated system to evaluate the compliance of the system with the corresponding requirements.
7. Regression Testing: Regression testing is done to make sure that the code changes do not affect the existing [functionality](#) and the features of the application. It concentrates on whether all parts are working or not.
8. Sanity Testing: Sanity testing is a subset of regression testing and is done to make sure that the code changes [introduced](#) are working as expected.
9. White box Testing: White box testing is a type of software testing that allows the tester to verify the internal workings [of](#) the software system. This includes analyzing the code, infrastructure, and integrations with the external system.
10. Black box Testing: Black box testing is a type of software testing where the functionality of the software system is tested without looking at the internal workings or structures of the software system.
11. Database Testing: Database testing is a type of software testing that checks the schema, tables etc., of the database under [test](#).
12. Adhoc Testing: Adhoc testing also known as monkey testing or random testing is a type of software testing where the test cases are selected arbitrarily and outcomes are compared to check whether they are correct.
13. Recovery Testing: Recovery testing is a type of software testing that verifies the software's ability to recover from failures like hardware failures, software failures, crashes, etc.
14. Static Testing: Static testing is a type of software testing that is performed to check the
15. defects in software without actually executing the code of the software application.
16. Grey-box Testing: Grey box testing is a type of software testing that includes black-box and white-box testing.
17. Component Testing: Component testing also known as program testing or module testing, done after the unit testing. In this, the test objects can be tested independently as a component without integrating with other components.

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#### Functional Testing Vs Non-Functional Testing

| Parameters       | Functional Testing                                                        | Non-functional Testing                                                      |
|------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Definition       | Functional testing verifies the operations and actions of an application. | Non-functional verifies the behavior of an application.                     |
| Testing based on | It is based on the requirements of the customer.                          | It is based on the expectations of the customer.                            |
| Objective        | The objective is to validate software actions.                            | The objective is to performance of the software system                      |
| Requirements     | Functional testing is carried out using the functional specification.     | Non-functional testing is carried out using the performance specifications. |
| Functionality    | It describes what the product does.                                       | It describes how the product works.                                         |
|                  | Unit testing.                                                             | Performance testing.                                                        |

|         |                                                                                 |                                                                           |
|---------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Example | Integration testing.<br>Sanity testing<br>Smoke testing.<br>Regression testing. | Load testing.<br>Stress testing.<br>Volume testing.<br>Usability testing. |
|---------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------|

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### Non-functional Testing

It is a type of software testing that checks the application for non- functional requirements like performance, scalability, portability, stress, etc. Various types of non-functional testing are Performance testing, Stress testing, Usability testing and so on.

#### Objectives of Non-functional Testing

- Increased usability: To increase usability, efficiency, maintainability, and portability of the product.
- Reduction in production risk: To help in the reduction of production risk related to non-functional aspects of the product.
- Cost Reduction: To help in the reduction of costs related to non-functional aspects of the product.
- Optimize installation: To optimize the installation, execution, and monitoring way of the product.
- Collect metrics: To collect and produce measurements and metrics for internal research and development.
- Enhance knowledge of product: To improve and enhance knowledge of the product behavior and technologies in use.